Six transponders · laid out like the network graph, each drawn as a copy of the single-transponder schematic

Every box is one copy of single_transponder_simple.pdf; the pulsers are replaced by the links. Wire colour = sending node. Blue monospace = variable names in toy_example.py / notebook N06.

i1 · pulser1 (U1)
sensor click at 1 ns
n = 2, same Ib2·split rule
t1
matrix row/col 2 · inputs: i1, t2, t3
ref_period = 3 ns
from i1
from t2
from t3
5 ns
5 ns
5 ns
R_in · 20 Ω
L_in · 5 nH
i_nw
bias3 in
I_b1 ≈ 150 µA (1.5 V via 10 kΩ)
Ib1 = 13 µA · transponder_bias1[2]
R2 · 4 Ω
R_T(t)
RT0 (+U2 when fired)
L1 · 5 nH
Ls1
R3 · 3 Ω
L_s1, R_s1
Rs1
Lb1 ≈ 148 nH
storage inductor (L_T)
LT
i_loop
U3
input nTron (Q1)
≈ L_nw1, R_nw1(t)
Inw1_sw
Lnw
Rnw ← Rnorm
d
g
s
① input loop
gate current into U3 = Σ amp·OutputPulse
② Q1 shunt loop
catches current
while U3 is resistive
③ storage loop
I_loop circulates; drains via R2;
drives U2 gate
bias4 in
I_b2 ≈ 160 µA (1.6 V via 10 kΩ)
Ib2 = 19 µA · transponder_bias2[2]
L2 · 5 nH
L_out
not modelled
U2
output nTron (Q2)
≈ L_nw2, R_nw2(t)
Inw2_sw
Rnorm (adds to RT)
④ output loop
U2 current pushed
into L2 / R1 → links
links →
split
fan-out n = 2
each link: Ib2·split/2
rest (1−split) → R1
R1 · 3 Ω
R_s2
not modelled
common ground (0 V)
t3
matrix row/col 4 · inputs: t1, t2
ref_period = 3 ns
from t1
from t2
Ib1 = 11.2 µA · transponder_bias1[4]
Ib2 = 19 µA · transponder_bias2[4]
split
fan-out n = 3
each link: Ib2·split/3
rest (1−split) → R1
t4
matrix row/col 5 · inputs: t3, t5, t6
ref_period = 3 ns
from t3
from t5
from t6
Ib1 = 12 µA · transponder_bias1[5]
Ib2 = 19 µA · transponder_bias2[5]
fan-out n = 2
each link: Ib2·split/2
t5
matrix row/col 6 · inputs: i1, t4, t6
ref_period = 3 ns
from i1
from t4
from t6
9 ns
5 ns
5 ns
Ib1 = 12 µA · transponder_bias1[6]
Ib2 = 19 µA · transponder_bias2[6]
fan-out n = 2
each link: Ib2·split/2
i2 · pulser2 (U4)
click at 1 ns + dt
n = 2, same Ib2·split rule
t2
matrix row/col 3 · inputs: i2, t1, t3
ref_period = 3 ns
from i2
from t1
from t3
Ib1 = 12 µA · transponder_bias1[3]
Ib2 = 19 µA · transponder_bias2[3]
fan-out n = 2
each link: Ib2·split/2
t6
matrix row/col 7 · inputs: i2, t4, t5
ref_period = 3 ns
from i2
from t4
from t5
Ib1 = 12 µA · transponder_bias1[7]
Ib2 = 19 µA · transponder_bias2[7]
fan-out n = 2
each link: Ib2·split/2

Legend

■ from i1 ■ from i2 ■ from t1 ■ from t2 ■ from t3 ■ from t4 ■ from t5 ■ from t6

Loops (same as the single-transponder drawing):

① input loop: gate current into U3, fed by the links (R_in/L_in) ② shunt loop: catches Ib1 while U3 is resistive

③ storage loop: I_loop circulates through Lb1 and R2 and drives the gate of U2 ④ output loop: U2 current pushed into L2/R1 and out on the links

Things to know:

· split enters at one place only: each output junction. Each link gets Ib2·split/n of nominal amplitude (peak ×0.61 after OutputPulse); the rest goes to the R1 shunt.

· The LTspice drawing has 2 inputs (R28/L18, R29/L19). The toy network needs 3 inputs for t1, t2, t4, t5, t6, so the input branch is repeated per link.

· LTspice values: Ib1 ≈ 150 µA, Ib2 ≈ 160 µA, R2 = 4 Ω, R3 = 3 Ω. Toy values: Ib1 = 12–13 µA, Ib2 = 19 µA, RT0 = 3 Ω, Rs1 = 2 Ω (notebook defaults).

· Sensors are drawn as the ideal pulsers of the original drawing, but the toy gives them the same output rule as a transponder (Ib2, split).

· L2, R1, L_nw2 and the R_in/L_in values are not variables in the toy; the delay boxes are the delay matrix entries; all ground rails are the same 0 V node.